# DANIEL I. STABILE

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## EDUCATION

**M.S. COMPUTER SCIENCE**
Cornell University, Bowers CIS
Ithaca, NY • Class of 2023
GEM Fellow at MIT Lincoln Laboratory

**B.S. COMPUTER SCIENCE**
Cornell University, College of Engineering
Ithaca, NY • Class of 2021
Magna Cum Laude • Ryan Scholar

## PUBLICATIONS

- **[Best Demo]** *An Adaptable, Safe, and Portable Robot-Assisted Feeding System* ACM/IEEE HRI 2024
- **[Best Paper Honorable Mention]** *Feel the Bite: Robot-Assisted Inside-Mouth Bite Transfer using Robust Mouth Perception and Physical Interaction-Aware Control* ACM/IEEE HRI 2024
- *TAACKIT: Track Annotation and Analytics with Continuous Knowledge Integration Tool* IEEE AIxDKE 2024

## EXPERIENCE

**Associate Technical Staff (Secret Clearance) | MIT Lincoln Laboratory | October 2023 - Present**

- Developed agentic mission planning application for generating drone flight plans using frontier LLMs (Anthropic, OpenAI) and Hugging Face's smolagents framework with multi-agent orchestration, structured outputs, built-in explainability, and layered validation
- Designed and implemented synthetic data generation pipeline and ML training/testing framework for anomaly detection in manufacturing cybersecurity applications; open-sourced select, non-sensitive datasets
- Fine-tuned YOLOv5 for maritime ship detection on electro-optical images, using Weights & Biases to systematically evaluate dataset quality and optimize training hyperparameters
- Engineered Python wrapper for NCAR's Weather Research and Forecasting (WRF) model on Lincoln Lab's HPC cluster to dynamically downscale multi-decadal global climate data to kilometer-scale resolution

**Graduate Researcher | EmPRISE Lab, Cornell University | January 2022 - May 2023**

- Built state-of-the-art robotic feeding system for people with severe mobility limitations, enabling safe and effective inside-mouth bite transfer; received best demo and honorable mention at HRI 2024
- Developed contact-aware compliant controller using ML classifiers (TCN, LSTM, SVM) trained on self-curated multimodal (visual + haptic) dataset to distinguish safe vs. unsafe physical interactions
- Used model predictive control to generate robot trajectories for feeding liquid food items that easily spill
- Designed and conducted human-robot interaction user studies for care recipients with mobility limitations to evaluate robot-assisted feeding system

**GEM Fellow Intern | MIT Lincoln Laboratory | Summer 2021**

- Built synthetic flight track generator to validate unsupervised ML classifiers for detecting anomalous airborne objects in US airspace

**Software Engineer Intern | Maculogix | Summer 2020**

- Reduced final product calibration time from 2.25 hours to 35 minutes while improving accuracy using advanced search algorithms, C#, OpenCV (Emgu), and FLIR Spinnaker SDK

## TEACHING & SERVICE

**Graduate Teaching Assistant | Cornell University | February 2020 - May 2023**

- CS 1110 - Intro to Computing in Python - Spring 2023, CS 4620 - Computer Graphics - Fall 2021 & Fall 2022 (Head TA), CS 4670 - Computer Vision - Spring 2022
- Led discussion sections and office hours and wrote exam questions; as Head TA, coordinated 10+ staff members on assignment creation and grading

**SoNIC Workshop 2022 | Cornell University | July 2022**

- Organized SoNIC, an intensive week-long robotics workshop in collaboration with Cornell Bowers CIS DEI Office, to inspire underrepresented students across the USA to pursue graduate studies in tech fields
- Created course content for theory and hands-on sessions, and instructed 30 participants to help develop "smart canes" – assistive technology to help the visually impaired navigate safely

## SKILLS

- **Coding** - Python, NumPy, W&B, SciKit-learn, PyTorch, JavaScript, TypeScript, C#, HTML & CSS, Bash
- **Tools** - Claude, HPC/SLURM, Jupyter Notebook, Git, Linux, Vim, VS Code, Docker
- **Design** - InDesign, Illustrator, Photoshop, Premiere Pro, After Effects, 3D Printing, Fusion 360